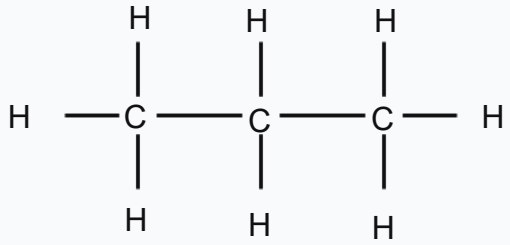
CARD 01 — PROPANE

Molecular Formula: C_3H_8 | IUPAC name: Propane

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_3H_8$$

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

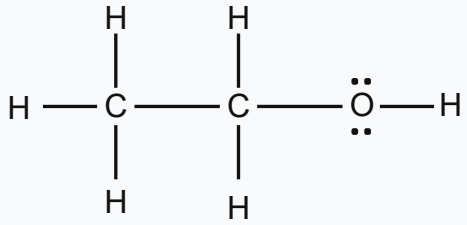
CARD 02 — ETHANOL

Molecular Formula: C_2H_6O | IUPAC name: Ethanol

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_2H_6O$$

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

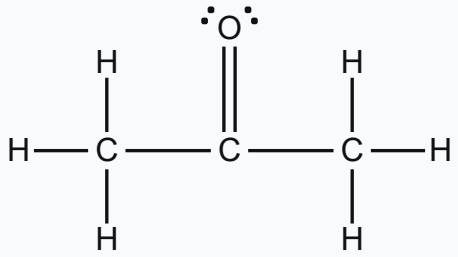
CARD 03 — ACETONE

Molecular Formula: C_3H_6O | IUPAC name: Propan-2-one

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_3H_6O$$

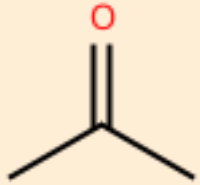
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

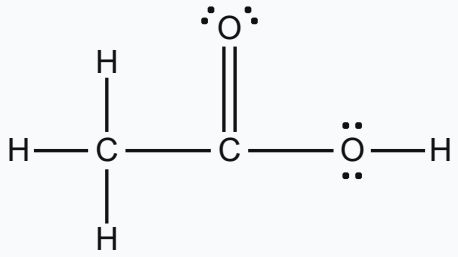
CARD 04 — ACETIC ACID

Molecular Formula: $C_2H_4O_2$ | IUPAC name: Ethanoic acid

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_2H_4O_2$$

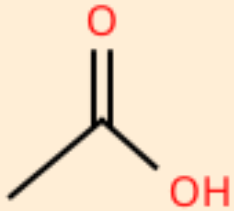
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

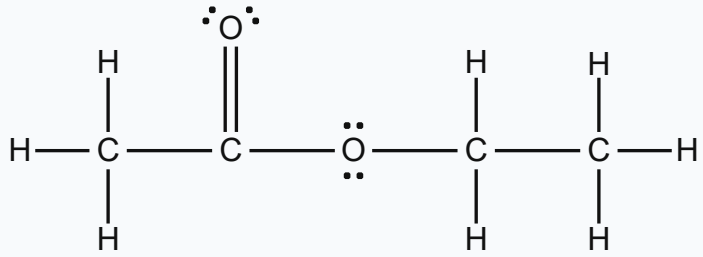
CARD 05 — ETHYL ACETATE

Molecular Formula: $C_4H_8O_2$ | IUPAC name: Ethyl ethanoate

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_4H_8O_2$$

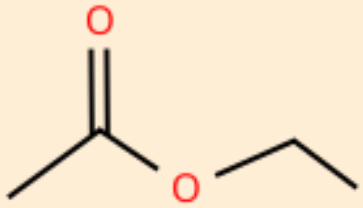
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)OCH_2CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

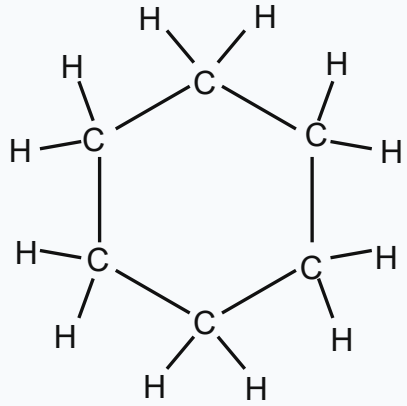
CARD 06 — CYCLOHEXANE

Molecular Formula: C_6H_{12} | IUPAC name: Cyclohexane

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_6H_{12}$$

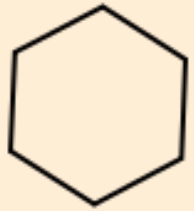
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6 \text{ (ring)}$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

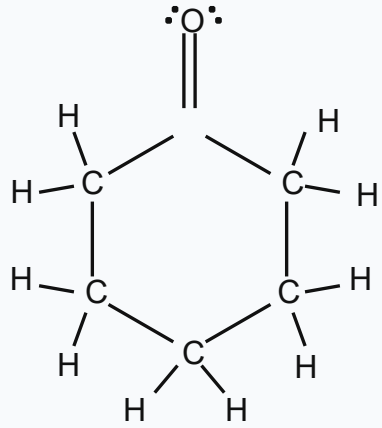
CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$ | IUPAC name: Cyclohexanone

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_6H_{10}O$$

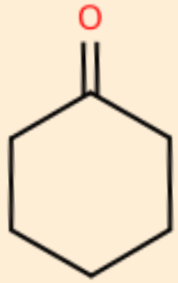
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5C{=}O \text{ (ring)}$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

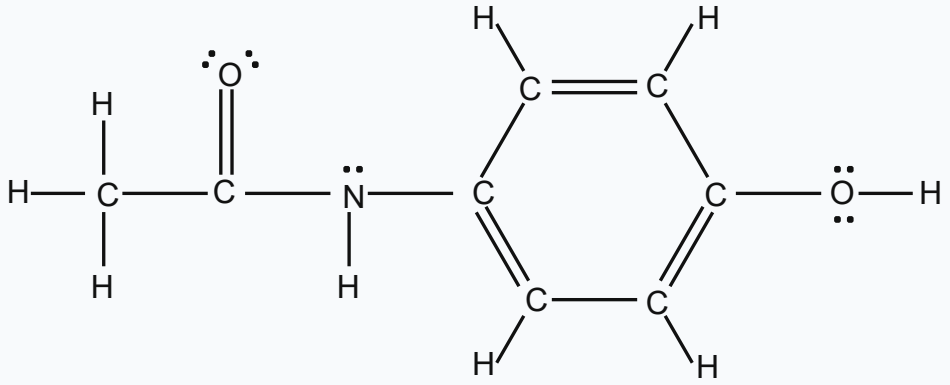
☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

CARD 08 — ACETAMINOPHEN (PARACETAMOL)

Molecular Formula: $C_8H_9NO_2$ | IUPAC name: N-(4-hydroxyphenyl)acetamide

1. MOLECULAR FORMULA / IDENTITY CHECK

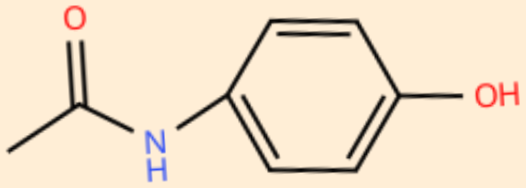
$$C_8H_9NO_2$$



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)NH–C_6H_4–OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

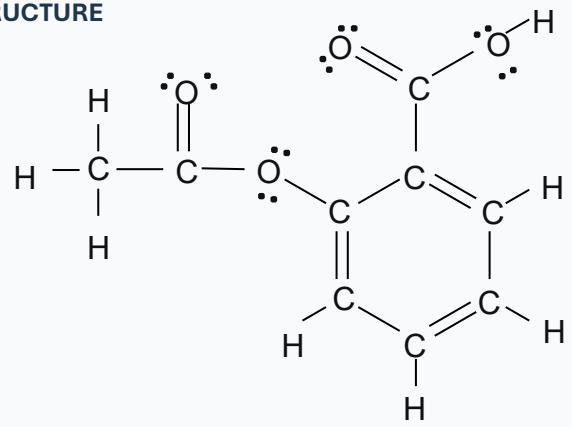
CARD 09 — ASPIRIN

Molecular Formula: $C_9H_8O_4$ | IUPAC name: 2-(Acetyloxy)benzoic acid

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_9H_8O_4$$

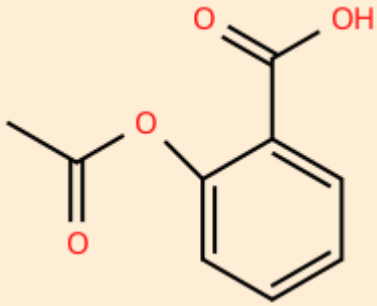
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)O–C_6H_4–C(=O)OH$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

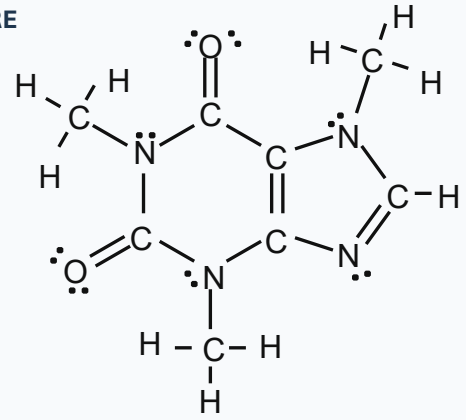
CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$ | IUPAC name: 1,3,7-trimethylpurine-2,6-dione

1. MOLECULAR FORMULA / IDENTITY CHECK

$$C_8H_{10}N_4O_2$$

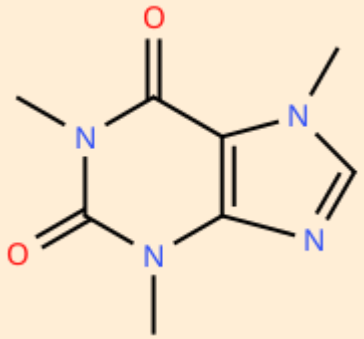
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3–N^1–CH=N–C^2=C^1–C(=O)–N(CH_3)–C(=O)–N^2–CH_3$$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed